Q1. Simulate 30 rolls with =RANDBETWEEN(1,6). What is the probability of rolling a 3 exactly 5 times? (Hint: Use BINOM.DIST)

A) This is a binomial probability problem because:

- Each roll of the die is a trial.
- Probability of success (rolling a 3) =(1/6)
- Number of trials = 30.
- We want the probability of exactly 5 successes The binomial probability formula is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Where:

- (n = 30) (trials)
- (k = 5) (successes)
- (p =1/6) (probability of rolling a 3)

In Excel, you can directly compute this with:

=BINOM.DIST(5, 30, 1/6, FALSE)

- `5` → number of successes
- `30` → number of trials
- `1/6` → probability of success
- `FALSE` → exact probability (not cumulative)

This will give you approximately:

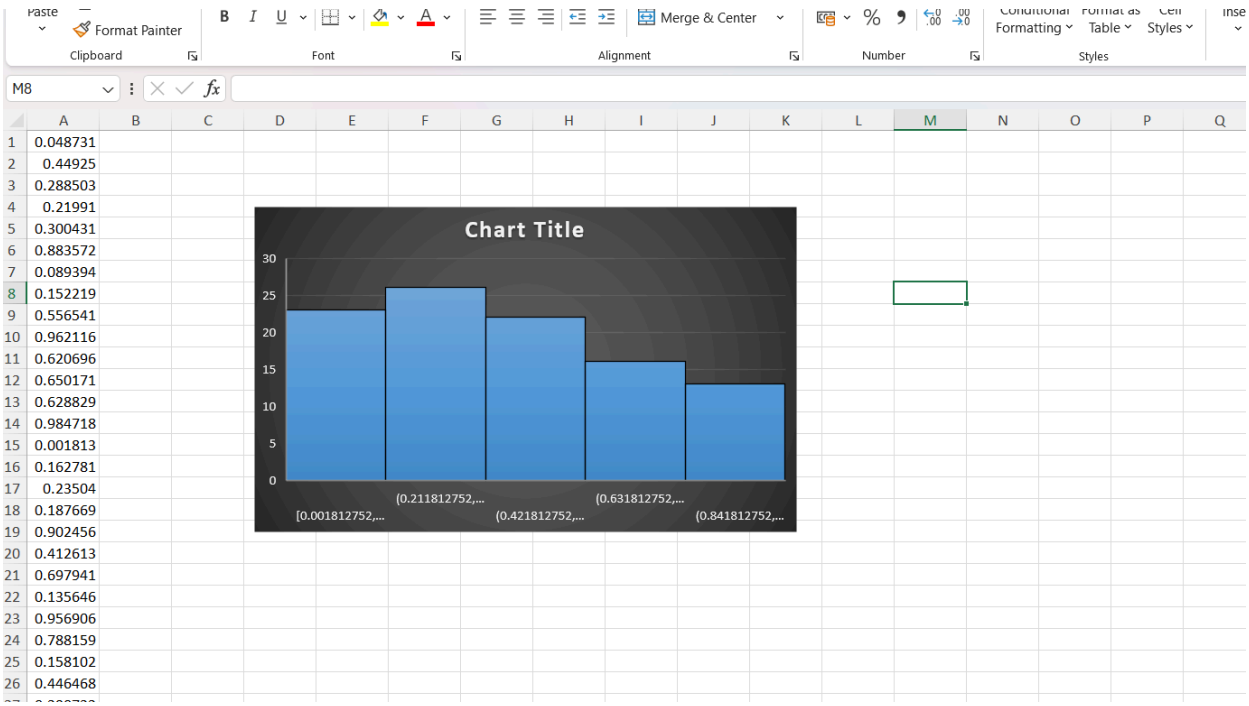
Probability of exactly 5 threes: 0.206

The probability of rolling a 3 exactly 5 times in 30 rolls is about 20.6%, and you can compute it in Excel using:

=BINOM.DIST(5, 30, 1/6, FALSE)

Q2. Generate 100 values in Excel using the continuous uniform distribution RAND() and plot a histogram. Describe the shape of the distribution.

A)



	A
1	0.048731
2	0.44925
3	0.288503
4	0.21991
5	0.300431
6	0.883572
7	0.089394
8	0.152219
9	0.556541
10	0.962116
11	0.620696
12	0.650171
13	0.628829
14	0.984718
15	0.001813
16	0.162781
17	0.23504
18	0.187669
19	0.902456
20	0.412613
21	0.697941
22	0.135646
23	0.956906
24	0.788159
25	0.158102
26	0.446468

- This generates random numbers between 0 and 1 from a continuous uniform distribution.
  - The histogram of 100 values generated with RAND() will show a uniform distribution — nearly flat across the interval [0,1], with no peaks or skewness.

Q3. A dataset has a mean of 50 and a standard deviation of 5. What percentage of values lie between 45 and 55 if the data follows a normal distribution?

A) Given Data:

- Mean (μ) = 50
- Standard deviation (σ) = 5
- Interval = [45, 55]

Convertto z-scores:

For x = 45:

```
z=x-μ  /  σ

z= 45 - 50 / 5 = -1
```

For x = 55:

```
z = 55 - 50 / 5 = +1
```

the probobility between z = -1 and z = +1.

Standard normaldistribution:

From z-tables (or the 68–95–99.7 rule):

- About 68.27% of values lie within ±1 standard deviation of the mean.

Approximately 68.3% of values liebitween 45 and 55.

Q4. What is the concept of standardization (z-score), and why is it important in data analysis? Explain the formula and how standardization transforms a dataset.

A) For a data point x:

z=x-μ / σ

- x = individual data value
- μ = mean of the dataset
- σ = standard deviation of the dataset

IT'S IMPORTANCE:

- Comparability: Different datasets may have different units or scales (e.g., test scores vs. salaries). Standardization puts them on a common scale.
- Outlier detection: Z-scores show how far a value is from the mean in terms of standard deviations. Large positive or negative z-scores indicate outliers.
- Statistical modeling: Many techniques (like regression, clustering, PCA, machine learning algorithms) perform better when variables are standardized.
- Probability interpretation: Z-scores link directly to the standard normal distribution, making it easy to calculate probabilities using z-tables or functions.

Standardization Transforms a Dataset:

- Original dataset: Mean = μ , Standard deviation = σ .
- After transformation:
- Mean becomes 0
- Standard deviation becomes 1
- Each data point is expressed as “how many standard deviations away from the mean” it is.

Example:

If a student’s score is 70, with μ =50 and σ =5:

z={70-50}/{5}=4

This means the score is 4 standard deviations above the mean, which is extremely high.

Q5. What is Kurtosis and their type?

A) Formula for Kurtosis:

For a data set with mean μ, standard deviation σ, and values x_i:

```
Kurtosis = Σ(x_i - μ)_4/n  /  σ_4
```

- It uses the fourth moment of the distribution.
- A normal distribution has a kurtosis of 3 (sometimes reported as 0 if "excess kurtosis" is used).

Types of Kurtosis:

- Mesokurtic
- Leptokurtic
- Platykurtic
- Value (Excess Kurtosis):
  - ≈ 0
  - 0
  - < 0
- Shape:
  - Normal bell curve
  - Taller peak, fatter tails
  - Flatter peak, thinner tails
- Interpretation:
  - Tails similar to normal distribution
  - More extreme values/outliers than normal

Q6. Explain why the uniform distribution is a good model for the outcome of rolling a fair die.

A)

Key Points

- Equal probabilities: In a fair die, the probability of rolling any number from 1 to 6 is the same, i.e., 1 / 6.
- Uniform distribution definition: A uniform distribution assigns equal probability to all possible outcomes within its range.

- Discrete uniform distribution: Since a die has a finite set of outcomes (1, 2, 3, 4, 5, 6), this is modeled by a discrete uniform distribution.

◆ Why It's Important

- Fairness assumption: Using the uniform distribution ensures that no number is favored, which matches the physical expectation of a fair die.
- Foundation for probability problems: Many probability exercises (like expected value, variance, or conditional probability) rely on the uniform distribution assumption for dice.
- Simulation and modeling: When simulating dice rolls in software or statistical analysis, the uniform distribution is the natural choice to replicate fairness.

Q7. Use Excel to compute the probability of getting at least 8 successes in 15 trials with success probability 0.5

A)

To calculate the probability of 'at least 8' successes ($P(X \geq 8)$), it's often easier to calculate the complement: 1 minus the probability of '7 or fewer' successes ($1 - P(X \leq 7)$).

The Excel formula for this would be:

```
=1 - BINOM.DIST(7, 15, 0.5, TRUE)
```

Explanation:

-BINOM.DIST(number_s, trials, probability_s, cumulative):

- number_s: The number of successes (7 in this case, because we want $P(X \leq 7)$).
- trials: The total number of trials (15).
- probability_s: The probability of success on each trial (0.5).
- cumulative: A logical value that determines the form of the function.
    - TRUE (or 1) for the cumulative distribution function ($P(X \leq \text{number_s})$).
    - FALSE (or 0) for the probability mass function ($P(X = \text{number_s})$).

By using TRUE for cumulative, the formula BINOM.DIST(7, 15, 0.5, TRUE) calculates the probability of getting 7 or fewer successes. Subtracting this from 1 gives you the probability of getting 8 or more successes.

Q8. How does log transformation help in stabilizing variance and making data more normally distributed?

A) Stabilizing Variance (Homoscedasticity):

Many statistical models assume that the variance of the errors (residuals) is constant across all levels of the independent variables. This is known as homoscedasticity. When this assumption is violated (heteroscedasticity), estimates can be inefficient, and hypothesis tests might be unreliable.

- How Log Transformation Helps: In datasets where the variance tends to increase with the mean (e.g., in count data or financial data), taking the logarithm of the data can compress the larger values more than the smaller values. This compression often reduces the spread of the data at higher values, leading to a more consistent variance across the range of the data. Essentially, it makes the spread of the data more uniform.

2. Making Data More Normally Distributed (Symmetry):

   Many statistical methods (like t-tests, ANOVA, and linear regression) assume that the data (or the residuals) are normally distributed. Positively skewed data (where the tail is on the right) is a common deviation from normality.

   - How Log Transformation Helps: Logarithms are particularly effective in handling positively skewed data. Because the logarithm function grows more slowly as its input increases, it effectively 'pulls in' the long right tail of a positively skewed distribution. This can make the distribution more symmetrical and closer to a normal distribution. For negatively skewed data, other transformations like squaring or cubing might be more appropriate.